HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Designed and implemented **industrial control strategies** using **Function Block Diagram (FBD) programming** for a new generation of cooling systems, integrating sensor and actuator data.
- Developed and tested **control software** for prototype data center cooling equipment, focusing on sequences of operation and variable-speed drive configurations.
- Configured and integrated **programmable controllers**, industrial sensors, and automation hardware onto prototype systems in an R&D; laboratory environment.
- Tuned **PID and cascaded control loops** for optimal performance of refrigeration and heat-rejection systems, achieving stable operation across varying load conditions.
- Supported testing, data collection, and analysis of control-system performance, identifying areas for improvement in efficiency and reliability.
- Authored comprehensive **software documentation**, managed version control using Git, and maintained detailed revision history for all control software developed.
- Containerized control system software using **Docker** and deployed to a testing environment on **AWS EC2**, ensuring consistency across development and R&D; lab setups.

SKILLS

Control Systems – Function Block Diagram (FBD), PID Tuning, Cascaded Control Loops, Equipment Sequencing, System Dynamics, HVAC Control, Refrigeration Control, Process Control, Industrial Automation

Programming Languages – Java, Python, SQL, R

Hardware & Controllers – Sensors, Actuators, Variable Speed Drives, Programmable Controllers, Automation Hardware

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Git, Jenkins

Data Analysis & BI – Power BI, Tableau, DAX, Power Query, pandas, NumPy, scikit-learn

Enterprise Systems – Spring Boot, REST APIs, PostgreSQL, MySQL, Oracle

Methodologies – Agile, Scrum

PROJECTS

Industrial Controller Software Development (HCL Technologies)

- Developed **Java Spring Boot applications** to interface with industrial automation hardware, managing data acquisition from sensors and controlling actuators.
- Implemented robust error handling and logging mechanisms using **Dynatrace** and **Splunk** for production monitoring and rapid issue resolution in an enterprise client environment.
- Created Python scripts to automate data extraction and pre-processing for control system performance analysis, reducing manual effort by approximately **30%**.

Prototype Cooling System Control Logic

- Designed and implemented **Function Block Diagram (FBD)** logic for a prototype data center cooling unit, controlling compressor stages and fan speeds based on temperature and pressure inputs.
- Tuned **PID loops** for precise temperature regulation, achieving target setpoints within **±0.5°C** and maintaining stable operation during simulated load changes.
- Integrated and tested control logic with various hardware components including **Carel controllers** and **variable-speed drives**, validating system response and safety interlocks.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)